15C 1 CARBOXYLIC ACIDS AND THEIR PHYSICAL PROPERTIES

SPECIFICATION REFERENCE
15.9 15.10

LEARNING OBJECTIVES

- Understand the nomenclature of carboxylic acids and be able to draw their structural, displayed and skeletal formulae.
- Understand that hydrogen bonding affects the physical properties of carboxylic acids, in relation to their boiling temperatures and solubility.

NOMENCLATURE

We met carboxylic compounds briefly in **Book 1**, as the products of the oxidation of primary alcohols.

The carboxylic acid group consists of a carbonyl group (C=O) with a hydroxyl group on the same carbon atom. Both groups influence each other, so carboxylic acids have some properties that are not shared with either carbonyl compounds or alcohols. The group is usually abbreviated to COOH (and sometimes to CO_2H, although this is not recommended).

The suffix *-oic acid* is used, with the carbon chain being named in the same way as for alkanes. If locants are used, then the carbon of the carboxylic acid group is carbon 1 in the chain. **Table A** shows some examples.

STRUCTURAL FORMULA	NAME	COMMON NAME
HCOOH	methanoic acid	formic acid
CH_3COOH	ethanoic acid	acetic acid
CH_3CH_2COOH	propanoic acid	(propionic acid)
$CH_3CH_2CH_2COOH$	butanoic acid	(butyric acid)
$(CH_3)_2CHCOOH$	methylpropanoic acid	(isobutyric acid)
HOOCCOOH	ethanedioic acid	oxalic acid
C_6H_5COOH	benzenecarboxylic acid	benzoic acid

table A

The general formula is RCOOH; the common names in brackets in **table A** are rarely used.

DISPLAYED AND SKELETAL FORMULAE

Table B shows the displayed and skeletal formulae of some carboxylic acids.

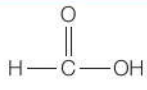
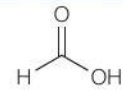
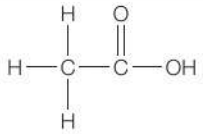
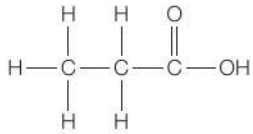
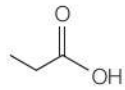
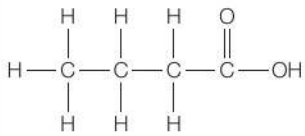
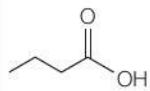
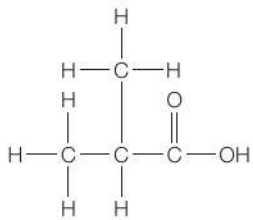
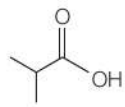
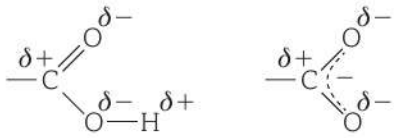
NAME	DISPLAYED FORMULA	SKELETAL FORMULA
methanoic acid		
ethanoic acid		
propanoic acid		
butanoic acid		
methylpropanoic acid		

table B

BONDING

You are familiar with the nature of the C=O double bond in carbonyl compounds, and of the single bonds in alcohols. The bonding in the carboxylic acid group is a combination of the two. There are now three polar bonds present.

Many reactions of carboxylic acids involve the loss of the hydrogen atom as H^+. This produces the carboxylate ion (COO^-) in which the two carbon–oxygen bonds are identical. **Fig A** compares the two.



carboxylic acid carboxylate ion

▲ **fig A** The structures of a carboxylic acid and its carboxylate ion.

Note that the charge and double bond character are evenly distributed across both oxygen atoms in the carboxylate ion.

PHYSICAL PROPERTIES

Carboxylic acids have distinctive smells and sour tastes. The smells of some of them are considered unpleasant. For example, butanoic acid is responsible for the smell of stale sweat, and hexanoic acid for the characteristic smell of goats. Citric acid gives lemons their very sour taste.

The Latin word for sour is *acidus*, and *acetum* means vinegar, the sour taste of which comes from the acetic acid it contains.

DID YOU KNOW?

Methanoic acid is found in ants and nettle stings. *Formica*, the origin of the name formic acid, is the Latin word for ant.



◀ **fig B** Methanoic acid, HCOOH, makes up more than half of an ant's body mass!

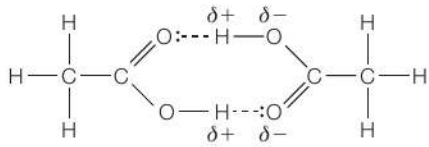
BOILING TEMPERATURES

The presence of three polar bonds, including a polar O—H bond, in the carboxylic acid group means that they have strong intermolecular forces (hydrogen bonding). This means that they have high boiling temperatures compared with other organic compounds with a similar molar mass. As the carbon chain lengthens, the London forces between the non-polar hydrocarbon chains increase. Therefore, boiling temperature increases with increasing molar mass, as shown in **table C**.

NAME	FORMULA	MOLAR MASS/ g mol^{-1}	BOILING TEMPERATURE /°C
methanoic acid	HCOOH	46	101
ethanoic acid	CH_3COOH	60	118
propanoic acid	CH_3CH_2COOH	74	141
butanoic acid	$CH_3CH_2CH_2COOH$	88	164

table C

The extent of hydrogen bonding in the shorter-chain carboxylic acids means that they form dimers (double molecules) in the absence of a solvent such as water. **Fig C** shows the hydrogen bonding (dashed lines) between two molecules of ethanoic acid.



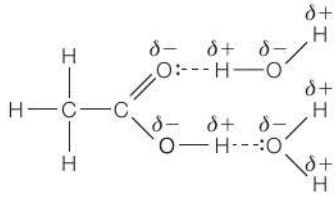
▲ **fig C** Hydrogen bonding in ethanoic acid.

EXAM HINT

In some forms of mass spectrometry, the ethanoic acid dimer can be seen at an *m/z* of 120.

SOLUBILITY IN WATER

The shorter-chain carboxylic acids are soluble in water because they can form hydrogen bonds with water molecules. **Fig D** shows hydrogen bonding between ethanoic acid and water.



▲ **fig D** Hydrogen bonding between ethanoic acid and water.

Solubility decreases with increasing chain length, as the hydrocarbon part of the molecules becomes larger.

LEARNING TIP

Note that the hydrogen bonding between carboxylic acid and water molecules involves:

–C=O ---- H–O–H and –O–H ---- O–H
|
OH

where hydrogen bonds are represented by dashed lines.

CHECKPOINT

1. Write the displayed formula for:
   (a) methanoic acid
   (b) ethanedioic acid.
2. Explain why hexanoic acid is much less soluble than ethanoic acid in water.

15C 2 PREPARATIONS AND REACTIONS OF CARBOXYLIC ACIDS

SPECIFICATION REFERENCE
15.11 15.12(i) 15.12(ii) 15.12(iii) 15.12(iv)

LEARNING OBJECTIVES

- Understand that carboxylic acids can be prepared by the oxidation of alcohols or aldehydes and the hydrolysis of nitriles.
- Understand the reactions of carboxylic acids with:
  (i) lithium tetrahydridoaluminate(III) (lithium aluminium hydride) in dry ether (ethoxyethane)
  (ii) bases to produce salts
  (iii) phosphorus(V) chloride (phosphorus pentachloride)
  (iv) alcohols in the presence of an acid catalyst.

PREPARATION BY OXIDATION

Although carboxylic acids are formed in many chemical reactions, two main methods are used to prepare them in the laboratory.

Oxidation uses either a primary alcohol or an aldehyde as the starting material. The usual oxidising agent is acidified potassium dichromate(VI), and the method is to heat the mixture under reflux. If a primary alcohol is used, it first oxidises to an aldehyde, then to a carboxylic acid, but both oxidations occur inside the apparatus used.

When the oxidation is complete, the reaction mixture is fractionally distilled to obtain a pure sample of the carboxylic acid.

EXAMPLES

Propanoic acid can be prepared from either propan-1-ol or propanal. The equations use [O] to represent the oxygen supplied by acidified potassium dichromate(VI).

From propan-1-ol:

$$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$$

From propanal:

$$CH_3CH_2CHO + [O] \rightarrow CH_3CH_2COOH$$

PREPARATION BY HYDROLYSIS

Nitriles are organic compounds containing the CN group. They can be hydrolysed by heating under reflux with either a dilute acid or aqueous alkali. The same apparatus is used for preparation and purification as for oxidation.

In both cases, the C≡N triple bond breaks. The carbon atom remains part of the organic product and the nitrogen atom becomes either ammonia or the ammonium ion.

ACIDIC HYDROLYSIS

Propanoic acid can be prepared from propanenitrile. The equation for the reaction is:

$$CH_3CH_2CN + H^+ + 2H_2O \rightarrow CH_3CH_2COOH + NH_4^+$$

ALKALINE HYDROLYSIS

Butanoic acid can be prepared from butanenitrile. The equation for the reaction is:

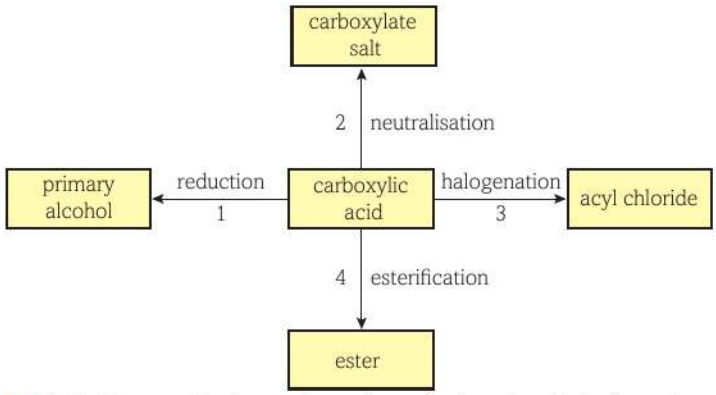
$$CH_3CH_2CH_2CN + OH^- + H_2O \rightarrow CH_3CH_2CH_2COO^- + NH_3$$

The product is actually the butanoate ion, but this is easily converted to butanoic acid by adding dilute acid:

$$CH_3CH_2CH_2COO^- + H^+ \rightarrow CH_3CH_2CH_2COOH$$

INTRODUCTION TO REACTIONS

There are four main reactions to consider. These are summarised in **fig A**.



▲ **fig A** These are the four main reactions of carboxylic acids to form other organic compounds.

Some of the products of these reactions, especially acyl chlorides and esters, will be discussed in more detail later on.

1 REDUCTION

You will remember that carboxylic acids are products of oxidation. They can be reduced in the same way as aldehydes and ketones. Carboxylic acids can be reduced to primary alcohols, but not to aldehydes. This is because aldehydes are more easily reduced than carboxylic acids, and so any aldehyde produced during the reduction will be immediately reduced to a primary alcohol. The reducing agent is lithium tetrahydridoaluminate, and is used in a solvent of dry ether. As an example, an equation using butanoic acid is:

$$CH_3CH_2CH_2COOH + 4[H] \rightarrow CH_3CH_2CH_2CH_2OH + H_2O$$

The product is butan-1-ol.

2 NEUTRALISATION

Although carboxylic acids are weak acids, they can be completely neutralised by mixing with aqueous alkali. The products are

carboxylate salts, which have a wide range of uses. The commonest, sodium ethanoate, is used in hand warmers and to make the additive that gives the flavour to salt and vinegar potato crisps. An equation for the reaction to make sodium ethanoate occurs between ethanoic acid and sodium hydroxide:

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

Carboxylate salts are ionic, and are sometimes shown with formulae such as $CH_3COO^-Na^+$. Their general formula is RCOONa.

3 HALOGENATION

In this reaction, the OH group is replaced by a halogen atom, usually chlorine, so the functional group becomes COCl. These products are known as acyl chlorides. They are highly reactive compounds with many uses in organic synthesis, as we will see later on. The reagent is phosphorus(V) chloride (old name 'phosphorus pentachloride'). It must be used in anhydrous conditions because both the reagent and the acyl chloride product react with water. The reaction is vigorous (very active), so no heating is required. An equation for the reaction, using propanoic acid as the example, is:

$$CH_3CH_2COOH + PCl_5 \rightarrow CH_3CH_2COCl + POCl_3 + HCl$$

Acyl chlorides are named using the suffix *-oyl chloride*, so this product is propanoyl chloride. Their general formula is RCOCl.

The phosphorus-containing product is phosphorus trichloride oxide (also known as phosphorus oxychloride). It is a liquid that mixes with the acyl chloride, which has therefore to be separated by fractional distillation. The hydrogen chloride gas produced escapes, appearing as misty fumes.

4 ESTERIFICATION

In this reaction, a carboxylic acid is mixed with an alcohol and a small amount of an acid catalyst, often concentrated sulfuric acid. Even with a catalyst, the reactions are slow and reversible.

Esters are used in industry for many purposes, including as solvents and in making polymers (polyesters). They also occur very commonly in nature – they are responsible for the characteristic smells of fruits, and most animal fats and vegetable oils are esters.

An equation for the reaction between methanoic acid and ethanol is:

$$HCOOH + CH_3CH_2OH \rightleftharpoons HCOOCH_2CH_3 + H_2O$$

The names of esters always contain two words – the first is the alkyl group from the alcohol and the second is the carboxylate from the carboxylic acid. So, this product is ethyl methanoate.

DID YOU KNOW?

Aspirin is a very familiar and inexpensive medication, and it is useful in treating many medical conditions. It is an example of a carboxylic acid.



fig B Aspirin tablets

LEARNING TIP

With an ester, be careful to make sure that the name and formula match. CH_3COOCH_3 is methyl ethanoate but $HCOOCH_2CH_3$ is ethyl methanoate.

CHECKPOINT

1. Write equations for the preparation of methylpropanoic acid:
(a) by the oxidation of an alcohol
(b) by the hydrolysis of a nitrile.

2. Write an equation for the conversion of methylpropanoic acid:
(a) into an acyl chloride
(b) into a methyl ester.

SKILLS DECISION-MAKING